LAUNCH THE PROJECT - 05

RGB Mood Lamp

(Intermediate level Project)

On AYNOP® UNO Launchpad Kit



Author: AYNOP Enterprises | Doc Version: 1.0 | Date: 11-04-2026

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1. Overview

This project demonstrates how to build a simple **RGB Mood Lamp** using the Arduino® UNO R4 Minima.

A **potentiometer** is used to control which colour of the RGB LED is turned ON. Only one colour (Red, Green, or Blue) is active at a time. By rotating the potentiometer knob, different colour zones are selected.

You will learn to:

- Read analog input using a potentiometer
- Control multiple digital outputs
- Divide analog input into different ranges
- Create simple interactive lighting effects

This project introduces the concept of **user-controlled lighting systems**, commonly used in smart lighting and decorative electronics.

2. Components Required

- Arduino® UNO R4 Minima board
- USB Type-C data cable
- RGB LED (Common Cathode)
- Potentiometer (10kΩ)
- 1 × 220Ω resistors (for RGB LED)
- Jumper wires
- Breadboard

3. Software Required

- Arduino IDE (v2.3.7 or later)

Note:
We assume the **Arduino UNO R4 Minima board package** is already installed on your machine, as explained in the 00_Getting_Started/00_GettingStarted_Arduino_R4_Minima guide. If it is not installed, please refer to that document and complete the installation before proceeding.

4. Hardware Setup

This section explains how to connect the RGB LED and potentiometer for mood lamp.

4.1 Wiring Diagram

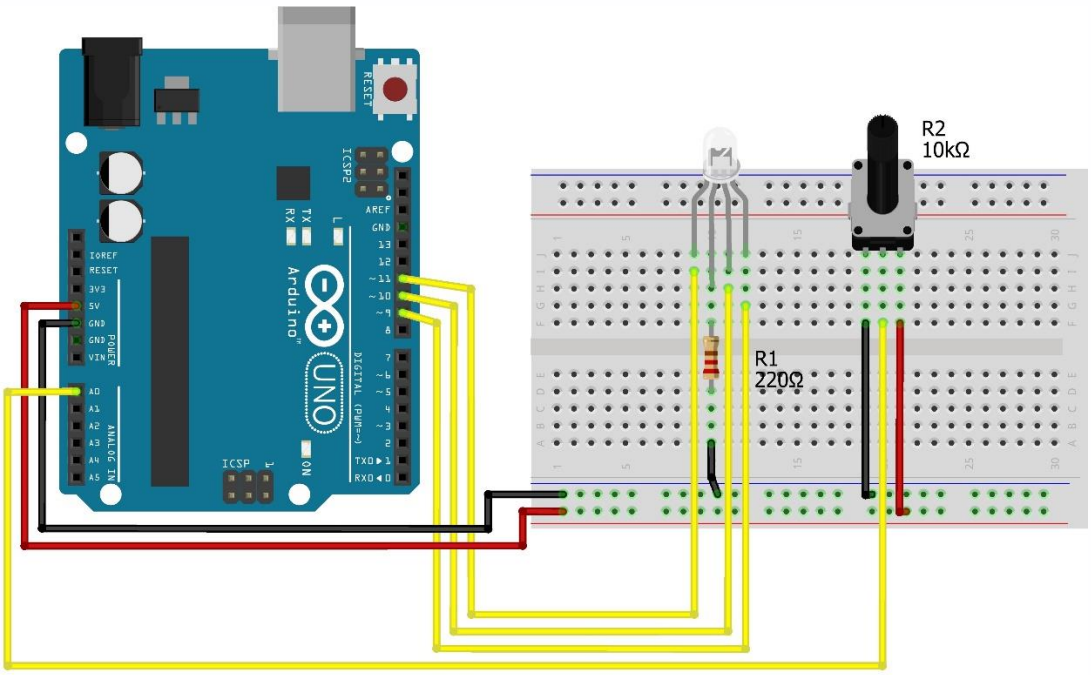


Figure 4.1 – Wiring diagram for RGB Mood Lamp

Tip: Always disconnect the USB cable before making or changing hardware connections.

5. Principle – How It Works

The RGB Mood Lamp is controlled using a **potentiometer**, which acts as a **variable voltage divider**. When the potentiometer knob is rotated:

- The output voltage changes smoothly between 0V and 5V
- The Arduino reads this voltage using analogRead()
- The value is converted into a digital range from 0 to 1023

Colour Selection Logic

The measured analog value is divided into **three distinct ranges**, each corresponding to a specific LED colour:

- **Low range (0 – ~348)** → Red LED ON
- **Middle range (~349 – ~813)** → Blue LED ON
- **High range (~814 – 1023)** → Green LED ON

At any given time:

• Only **one colour is activated**
• The other two colours remain OFF

This creates a clear and stable colour transition based on the knob position.

6. Procedure – Steps to Run

1. **Build the Circuit**
   a. Connect the RGB LED and Potentiometer as shown in the wiring diagram (Figure 4.1).

2. **Connect the Board**
   a. Use a USB Type-C data cable to connect your UNO R4 Minima to your computer.

3. **Open the Project Code**
   a. Simply **double-click** the file *05_RGB_Mood_Lamp.ino* in the project folder, and it will open directly in the Arduino IDE (if installed).

4. **Confirm Board Selection**
   a. The IDE usually auto-detects the UNO R4 Minima if the package is installed.
   b. If not installed, refer to the *00_Getting_Started/00_GettingStarted_Arduino_R4_Minima* document to install the necessary board package.
   c. Verify that *Arduino UNO R4 Minima* is selected in the IDE's board selector (top toolbar).

5. **Upload the Code**
   a. Click the **Upload** button (arrow icon) in the top-left corner of the IDE.
   b. Wait until the console displays "**Done uploading**."

6. **Observe the Behaviour**
   a. Rotate the potentiometer knob slowly.
   b. The RGB LED changes colour based on the knob position.
   c. Only one colour is active at a time.

7. Expected Output

- **Potentiometer Low Position:**
  - Red LED turns ON.
- **Potentiometer Middle Position:**
  - Blue LED turns ON.
- **Potentiometer High Position:**
  - Green LED turns ON.

💡 **Tip:** Rotate the potentiometer slowly and observe how the colour changes in steps based on position.

8. Code

The source code for this project is included in the downloaded folder:

📂 uno-launchpad-kit/02_Intermediate_Projects/ 05_RGB_Mood_Lamp/ 05_RGB_Mood_Lamp.ino

👉 **Tip:**

- To open the project, simply **double-click the .ino file**. If the Arduino IDE is installed, it will launch automatically and load the code.
- If you **haven't installed the Arduino IDE yet**, please refer to:
  📂 uno-launchpad-kit/00_Getting_Started/00_GettingStarted_Arduino_R4_Minima
  to **download and install it.**

8.1 Function References

- setup() – Runs once when the board is powered on or reset. Used to initialize sensors, LCD, and I/O pins.
- loop() – Runs continuously after setup () finishes. Reads sensor values and updates the LCD at fixed intervals.
- analogRead(pin) – Reads analog voltage from the potentiometer.
- digitalWrite(pin, value) – Turns LED pins ON or OFF.
- pinMode(pin, mode) – Sets pins as INPUT or OUTPUT.
- delay(ms) – Adds a small delay for smooth servo motion.

📚 **For more details and advanced usage, visit:**

🔗 Arduino Language Reference — The official guide for all Arduino functions.

9. Troubleshooting Tips

- **LED not lighting up?**
  - Check polarity of RGB LED.
  - Ensure common pin is connected to GND via 220 Ohm resistor.

- **Only one colour working?**
  - Check wiring of other LED pins (D9, D10, D11).
  - Ensure resistors are properly connected.

- **Colours not changing with potentiometer?**
  - Check potentiometer wiring.

- **Wrong colour sequence?**
  - Verify correct pin mapping for RGB LED.
  - Check if LED is common cathode type.

- **Still not working?**
  - Recheck all wiring connections carefully.
  - Disconnect power and rebuild the circuit step-by-step.
  - Test each module (RGB & Potentiometer) individually with help of basic projects.

10. License

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11. Support & Feedback

We value your feedback and are happy to assist with any questions, troubleshooting, or suggestions you may have.

Email: *support@aynop.com*

When sending an email, please include your kit name (AYNOP® UNO Launchpad Kit) and, if applicable, the project name in the subject line. This will help our team respond faster and more accurately. We aim to respond to all queries within 2–3 business days. Your feedback helps us improve our products and create even better learning experiences for all Arduino beginners.